



TASK REPORT -WEEK-2

1. Vulnerability Scanning

In this task, a basic vulnerability assessment was performed on the target system using Nmap for service and port scanning, Nikto for web server vulnerability detection, and OpenVAS for comprehensive vulnerability scanning and risk analysis. The objective was to identify security weaknesses such as misconfigurations, outdated software, and exposed services. The results were analyzed and documented with severity and risk levels to understand the security posture of the target system and highlight the importance of applying proper security controls.

After running the scan below are the results,

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Session Actions Edit View Help
Host is up (0.0000030s latency).
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 256 IP addresses (4 hosts up) scanned in 7.20 seconds

[nighthacker@Hacker]~$ nmap -sV -O 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-05 10:20 IST
Nmap scan report for 10.0.2.3
Host is up (0.0022s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshexec
513/tcp   open  login
514/tcp   open  tcpwrapped
1899/tcp  open  java-rmi       GNU Classpath gmicregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8080/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:00:1E:5B7 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.93 seconds

[nighthacker@Hacker]~$
```

Fig1



Here is the table showing results,

Scan ID	Vulnerability	CVSS Score	Priority	Host
SCN-01	VSFTPD 2.3.4 Backdoor Command Execution	9.8	Critical	10.0.2.3
SCN-02	OpenSSH 4.7p1 User Enumeration / Weak Crypto	6.8	Medium	10.0.2.3
SCN-03	Telnet Service Enabled (Cleartext Credentials)	8.2	High	10.0.2.3
SCN-04	Bind Shell Exposure on Port 1524	10.0	Critical	10.0.2.3
SCN-05	Samba 3.x Remote Code Execution (CVE-2007- 2447)	10.0	Critical	10.0.2.3
SCN-06	Apache 2.2.8 Multiple Vulnerabilities (Outdated Web Server)	7.5	High	10.0.2.3
SCN-07	MySQL 5.0.51a Weak Default Credentials	8.5	High	10.0.2.3
SCN-08	PostgreSQL Open Remote Access	7.2	High	10.0.2.3
SCN-09	VNC Service Without Strong Authentication	9.0	Critical	10.0.2.3
SCN-10	Java RMI Registry Remote	8.6	High	10.0.2.3



	Exploitation Risk			
SCN-11	NFS Misconfiguration (Unauthorized Mount)	7.8	High	10.0.2.3
SCN-12	Tomcat 8080 Default Credentials / Manager Exposure	8.8	Critical	10.0.2.3
SCN-13	IRC UnrealIRCd Backdoor Vulnerability	10.0	Critical	10.0.2.3
SCN-14	ProFTPD 1.3.1 Known Remote Vulnerabilities	8.1	High	10.0.2.3